

Medical Question Answering

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Abstract

Question answering systems (QAS) have significant potential for supporting clinical and research decision making in specialised domains such as biomedicine. We design a QAS system using the PubMedQA dataset and systematically evaluate how **text representation** and **prediction method** affect **answer quality**. We compare four systems against a TF-IDF and logistic regression baseline: (1) frozen Biomed-BERT embeddings with logistic regression, (2) end-to-end fine-tuned BioClinical Modern-BERT with a classification head, (3) BioGPT fine-tuned for generative answer-prediction, and (4) BioMed-R1-8B, a reasoning-oriented large language model. Results show that denser biomedical contextual representations outperform sparse TF-IDF features, and that task-aligned model design matters more than model complexity. BioMed-R1-8B achieved the highest accuracy at **78.20%**, BioClinical Modern-BERT provided the strongest practical discriminative classifier with a macro-average F1 score of **0.56**, while BioGPT best handled uncertainty, with maybe-class F1 at **0.41**. Across all systems, the 'maybe' class demonstrated the most consistent failure mode, revealing a persistent challenge in modelling uncertainty. Additional answer-quality prediction analysis showed that reasoning-related question features were a key determinant of question difficulty. Future improvements should address the challenge of reasoning-heavy instances with a multi-step system that escalates to stronger reasoning models for low-confidence instances.

1 Introduction

Question answering systems (QAS) are a central goal of Natural Language Processing, aiming to provide precise answers in natural language. Such systems can enable efficient access to large-scale information, significantly reducing the time and effort required compared to a manual search. This is particularly relevant in rapidly expanding,

highly specialised domains such as biomedical research, where accessing accurate and relevant information can directly influence critical decisions made by clinicians and researchers. As a result, domain-specific QAS provide a useful tool to support evidence-based decision making by facilitating faster and more reliable information retrieval.

In our project, we aim to design and implement a QAS capable of answering biomedical questions using the PubMedQA dataset (Jin et al., 2019). Our primary objective is to investigate how different text representations and model architectures affect the quality of answers. Traditional bag-of-words and TF-IDF text representations are simple and easy to interpret but often fail to capture semantic relationships and contextual meaning. In contrast, more advanced embedding-based methods can represent text in a way that better captures meaning and context, which is particularly important in specialised domains. However, more complex methods are often associated with higher cost and resource demand, are less interpretable and are harder to debug. Simpler methods are therefore worth exploring. Our second key design consideration is model prediction method. Classification approaches are typically more efficient, more consistent, easier to train, and performance is easier to evaluate, especially when answers are limited to predetermined labels. However, they may struggle to capture nuance. In comparison, generative models are more adaptable and often demonstrate higher accuracy but provide additional challenges such as increased complexity, potential for error and hallucinations, higher inference cost and increased training costs. Understanding how text representation and model architecture interact to affect answer quality is key to optimising performance. By systematically comparing alternative approaches, we aim to identify which methods are most effective for handling the challenges of biomedical question answering.

Question:

Do preoperative statins reduce atrial fibrillation after coronary artery bypass grafting?

Context:

(Objective) Recent studies have demonstrated that statins have pleiotropic effects, including anti-inflammatory effects and atrial fibrillation (AF) preventive effects [...]

(Methods) 221 patients underwent CABG in our hospital from 2004 to 2007. 14 patients with preoperative AF and 4 patients with concomitant valve surgery [...]

(Results) The overall incidence of postoperative AF was 26%. Postoperative AF was significantly lower in the Statin group compared with the non-statin group (16% versus 33%, $p=0.005$). Multivariate analysis demonstrated that independent predictors of AF [...]

Long Answer:

(Conclusion) Our study indicated that preoperative statin therapy seems to reduce AF development after CABG.

Answer:

Yes

Figure 1: An instance (Sakamoto et al., 2011) from PubMedQA dataset (Jin et al., 2019).

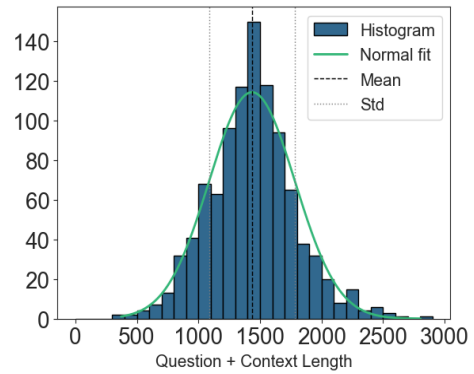
We use the PubMedQA dataset which is constructed from biomedical research papers. Instances consist of a question derived from a research title, a corresponding abstract (excluding the conclusion) that provides context, a long answer (the abstract conclusion) and a yes/no/maybe short answer (Figure 1). This type of biomedical data can pose significant challenges. It often contains complex terminology, domain-specific phrases and abbreviations which can be difficult for models to interpret and may require reasoning rather than simple keyword matching. Additionally, abstracts are usually dense and information-rich, requiring models to identify and extract relevant details from long passages of text. This is highlighted by Figure 2a which demonstrates an average input length of 1450 words, with some inputs exceeding 2500 words. The dataset demonstrates a significant label imbalance with 55.2% ‘yes’ class, 33.8% ‘no’ class and only 11% ‘maybe’ labels (Figure 2b). Accuracy alone will therefore be an insufficient metric and macro-average F1 will provide a more reliable measure of performance by weighting each class equally. These challenges highlight the importance of selecting appropriate text representations and prediction models to achieve high accuracy and reliability.

2 Methods

2.1 Baseline System

The baseline system (Figure 3) follows a six stage pipeline. The input, which consists of a ques-

(a) Combined Question and Context Length Distribution



(b) Label Distribution

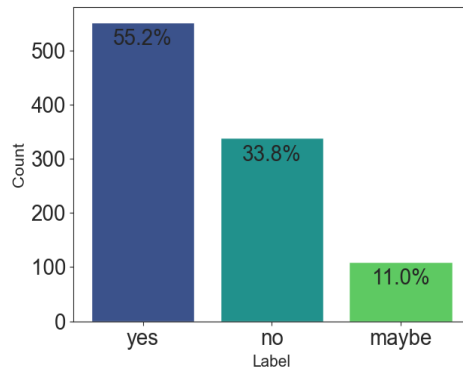


Figure 2: Data Exploration Plots showing the distribution of input length and label class split.

tion paired with a corresponding abstract, is preprocessed by the vectorization stage which transforms the text using scikit-learn’s TfidfVectorizer. This is configured with English stop word removal and bigram feature extraction, which results in tokens that are lowercased, split into unigrams and bigrams, and filtered using English stop-words capturing both individual terms and short phrase patterns. This is particularly important in biomedical question answering, where discriminative features often appear in multi-word expressions. The TF-IDF representation assigns weight to terms based on their frequency within a document and their rarity across the corpus. The weight of a term (t) in the document (d) is computed as:

$$TF-IDF(t, d) = tf(t, d) \cdot \log(N/df(t))$$

Where $tf(t, d)$ is the frequency of t in the document, N is the total number of documents, and $df(t)$ is the number of documents containing t . This yields a high-dimensional sparse vector representation for each input, emphasising discriminate terms while down-weighting common language terms. For prediction, feature vectors represent-

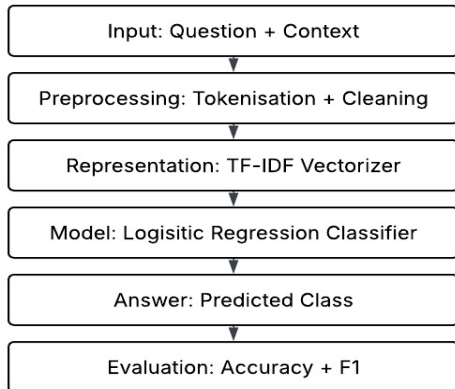


Figure 3: Baseline System showing the six stage pipeline consisting of input, preprocessing, text representation, prediction model, produced answer and evaluation.

ing question-context pairs are passed into a logistic regression classifier which uses regularised cross-entropy loss. The model produces a weight vector that maps sparse TF-IDF features to class probabilities over the three output classes. Model output is evaluated per class using accuracy and macro-averaged F1 score, ensuring that performance is not dominated by the majority class.

2.2 Axes of Experimentation

In order to evaluate performance and answer quality, we systematically experiment with variations in text representation and prediction method, comparing sparse text representations with denser embeddings and classification methods with answer generation, respectively. These two components are most influential in overall system performance, and improvements in these axes are hypothesized to provide significant performance gains. Models must reason over long, information-dense contexts, which will be particularly challenging for sparse representations that struggle to capture long-range dependencies. The baseline system using TF-IDF text representation is expected to perform adequately on 'reasoning-free' instances but will struggle with complex questions and abstracts. TF-IDF representations are not able to capture word order and context which limits their ability to capture nuanced meanings and semantic relationships between words. The second key component of our QA pipeline is the prediction method. The baseline system uses a logistic regression classifier which was chosen for its simplicity, interpretabil-

ity, cost and time efficiency on high-dimensional sparse data. However, its performance is limited by its linear nature which restricts its ability to capture complex patterns and interactions within non linear data. In order to improve on the limitations of our baseline system we systematically test improvements across the text representation and model axes and evaluate effects on performance (Table 1).

2.3 Proposed Improvements

We hypothesise that increasing complexity across text representation and prediction method will lead to performance improvement. Jin et al. (2019) achieves a single-human performance accuracy of 78% on reasoning-required instances (Table 2) providing a reference point to aim for. All models are evaluated using **accuracy**, **macro F1**, and **per-class F1**. Macro F1 is used as a primary indicator of class imbalance handling (Younes and Mathiak, 2022).

2.3.1 Improvement 1: BiomedBERT Embeddings and Logistic Regression

Our first experiment retains the cheap, simple, and quick linear logistic regression classifier to establish performance gained by using **dense biomedical contextual embeddings** instead of sparse TF-IDF features. This allows us to explore the limits and bottleneck of text analytics using linear classifiers, and to directly isolate the effect of the representation axis. We select **BiomedBERT** which is pretrained from scratch using domain-specific biomedical corpora and outperforms continuous pre-training of general models in biomedical NLP tasks. Gu et al. (2021) shows a fully fine-tuned BiomedBERT applied to PubmedQA dataset achieving 55.84% accuracy. This provides a reference point for expected improvement of our improved frozen encoder + linear classifier strategy.

2.3.2 Improvement 2: BioClinical ModernBERT

Our second experiment builds directly on the first by replacing both components with **end-to-end classification** using **BioClinical ModernBERT** with a sequence-classification head, so that text representation and prediction are learned jointly within a single neural architecture. The paired question-context input is processed by the transformer encoder to produce contextualised to-

	Text Representation	Prediction Method	Model
Baseline	TF-IDF	Linear classification	Logistic Regression
Experiment 1	BiomedBERT embeddings	Linear classification	Logistic Regression
Experiment 2	Contextual transformer representation	End-to-end classification	BioClinical ModernBERT
Experiment 3	Generative biomedical representation	Answer generation	BioGPT
Experiment 4	Generative biomedical representation	Answer generation (Reasoning model)	Biomed-R1-8B

Table 1: Experimental pairings across Text Representation, Prediction Method, and Model.

ken embeddings that capture semantic relationships, long-range dependencies, and biomedical language more flexibly than a fixed embedding space. The final sequence representation ([CLS]) is passed to a fully connected layer followed by a SoftMax function to produce class probabilities:

$$P(y = c \mid x) = \text{SoftMax}(Wh_{cls} + b)$$

where h_{cls} is the contextual sequence representation and W and b are the classification-head parameters. This end-to-end setup allows the model to learn both the representation and the decision boundary jointly via backpropagation. We therefore expect it to perform better on examples where correct prediction depends on subtle biomedical phrasing, paraphrase, or complex interactions between the question and abstract (Gu et al., 2021; Warner et al., 2025).

2.3.3 Improvement 3: BioGPT

Our third proposed improvement utilises **BioGPT**, a domain-specific generative Transformer model pre-trained on large-scale biomedical literature, designed to both understand and generate medically relevant natural language. While Experiments 1 and 2 focus on the enhancement of text representation, this experiment instead focuses on the second axis of experimentation. We replace the classification-based approach with BioGPT, a decoder-only transformer model that performs autoregressive answer generation. Importantly, this experiment is independent of text representation, as BioGPT operates on transformer-based contextual representations. The tokenised question+context input is processed to produce contextual representations, and the model predicts the next token sequentially to generate the final answer. This setup enables a direct comparison between classification

and generation under a consistent representation paradigm, isolating the effect of the prediction method. BioGPT has produced successful results in the literature and we therefore expect this generative approach to better capture complex biomedical relationships and demonstrate improved performance on reasoning-required instances (Luo et al., 2022).

2.3.4 Improvement 4: Biomed-R1-8B

Our final proposed improvement utilises **BioMed-R1-8B** which is a specialized **large language model** that emulates the medical reasoning process. BioMed-R1-8B is trained specifically to reason, producing an intermediate chain-of-thought before committing to a final answer. This is expected to result in enhanced performance on reasoning-heavy instances compared to BioGPT. It is developed using a combination of supervised fine-tuning(SFT) and reinforcement learning (RL) specifically focused on "reasoning heavy" examples which has been shown to outperform SFT-only and RL-only approaches (Thapa et al., 2026). Previous studies comparing open-source biomedical reasoning models BioMed-R1-8B achieve 63.3% knowledge accuracy and 49.2% reasoning accuracy (Thapa et al., 2026). We therefore expect this model to demonstrate improved handling of reasoning-required instances and improved performance overall.

3 Experimental Evaluation

3.1 Experimental Setup

Models are evaluated using accuracy and macro-average F1 score. F1 score is used in addition to accuracy due to uneven class distribution resulting in model bias towards the majority class. All experiments use the expert-labelled subset of the Pub-

	Accuracy	F1
Majority-class		
Baseline	0.552000	0.237113
Reasoning-free		
Human Performance	0.904000	0.841823
Reasoning-required		
Human Performance	0.780000	0.721920

Table 2: Majority-class Baseline and Human Performance for reasoning-free and reasoning-required instances (Jin et al., 2019). Majority-class Baseline describes the accuracy that would be achieved if all answers were predicted as ‘yes’.

MedQA dataset, PQA-L, which contains 1000 instances. Following the standard protocol (Jin et al., 2019) the dataset is split into a 500-instance cross-validation split, with the remaining 500 instances reserved for the test set.

Our experimental setup starts with a TF-IDF + logistic regression baseline, followed by BiomedBERT embeddings with the same classifier to isolate the effect of denser text representations. The final classification model replaces both components with BioClinical ModernBERT, enabling end-to-end learning. Finally, the BioClinical ModernBERT classification model is compared with BioGPT and BioMed-R1-8b to contrast discriminative, generative and reasoning based generative approaches respectively.

3.1.1 Baseline Experiment

In the baseline experiment, each instance is represented by concatenation of the question and its associated context and mapping categorical labels (‘yes’, ‘no’, ‘maybe’) to numeric classes. Concatenation of the inputs allows the task to focus on comprehension and reasoning of the text, instead of retrieval, enabling easier evaluation of model performance. Text features are extracted using a **TF-IDF vectorizer** with English stop-word removal and unigram-bigram representations, fitted on the aggregated 500-instance cross-validation split and applied to the held-out test set. A **multinomial logistic regression classifier** with balanced class weights and a maximum of 1000 iterations is trained on the complete feature matrix and used to generate predictions and class probabilities for the test data.

3.1.2 Experiment 1: BiomedBERT Embeddings and Logistic Regression

Our first experiment uses **BiomedBERT** to encode the question and context separately. BiomedBERT is pre-trained on PubMed abstracts and full-text articles making it well suited to handle the complexities of biomedical language. Four vector combination methods to merge question and context embeddings are tested:

- **Concat**: Concatenation, preserving both representations independently.
- **Diff**: Element-wise subtraction, capturing semantic distance between question and context.
- **Prod**: Element-wise multiplication, highlighting dimensions where both vectors agree.
- **Concat-Diff**: Concatenation of both the original vectors and their difference.

All preprocessing is done by the tokenizer of BiomedBERT. Seven parameter configurations were tested: token length allocation (question length: MAX-Q, context length: MAX-C), pooling method, context length limit (CTX-LIMIT), regularisation strength (C), class-weight and batch-size. Parameters are configured as follows: questions are truncated to a maximum of 128 tokens (MAX-Q = 128), and the contexts are truncated to a maximum of 384 tokens (MAX-C = 384), ensuring that the total number of tokens for both remains within the 512-token input limit of BiomedBERT. Before tokenisation, each context retains only the first three paragraphs (CTX-LIMIT = 3). This was found to effectively reduce noise introduced by subsequent paragraphs while retaining the most relevant information. Subsequently, the token outputs at all positions are subjected to mean pooling to generate a 768-dimensional sentence vector for each input, rather than relying solely on the representation of the [CLS] token. All embedding features are standardised to zero mean and unit variance using scikit-learn’s StandardScaler before applying a logistic regression classifier with the same settings as baseline (class weight=‘balanced’).

3.1.3 Experiment 2: BioClinical ModernBERT

Experiment 2 extends Experiment 1 by replacing frozen BiomedBERT embeddings plus logistic regression with **end-to-end BioClinical ModernBERT classification**, so the representation and classifier are learned jointly rather than in a fixed embedding space.

Inputs use paired question and contexts with truncation applied only to the context to retain full question length. Maximum length was set to 512, which covered almost all analysed expert examples while remaining computationally practical. Implementation used Hugging Face transformers and datasets; supporting hyperparameters are listed in Appendix 5.

Early supervised runs collapsed on 'maybe', so class imbalance became the main optimisation goal. Comparing plain fine-tuning, class-weighted loss, weighted loss plus weighted sampling, and weighted sampling showed that weighted sampling best stabilised minority-class learning. We also tested masked language model adaptation on unlabelled PQA-U but performance was not significantly improved so this was not further explored (Gururangan et al., 2020). We then warm-started the encoder on ~200k weakly labelled binary yes/no examples from PQA-A. A 10% pilot on this weak data fixed the final warm-start recipe, weighted sampling and learning rate 1×10^{-5} ; the best full-data checkpoint was then used for downstream fine-tuning. Final fine-tuning used the 1000 PQA-L split: 500 development, 500 test, and outer 10-fold cross-validation on the dev set. Each fold: 400 training, 50 calibration and 50 evaluation examples; separating fitting, calibration, and evaluation.

Two pipelines were compared. **Pipeline-S** used the weak warm-start, a fresh three-class head, supervised fine-tuning on the 400 training examples, temperature scaling on the 50-example calibration split to soften overconfident probabilities (Guo et al., 2017), and evaluation on the outer 50. **Pipeline-E** added semi-supervised learning: two teachers with different random seeds were trained on the same 400 examples to provide diversity for agreement-based pseudo-label filtering; both were calibrated, used to score the full PQA-U pool, and filtered with class-aware confidence and margin thresholds, after which the highest-ranked soft pseudo-labels were downweighted and used alongside the expert data to train a student. The winning pipeline was chosen by mean outer-fold macro F1, then evaluated once on the untouched 500-example test set using a calibrated 10-member soft-voting committee, which preserves probability information and reduces variance across members.

3.1.4 Experiment 3: BioGPT

Our third experiment replaces answer classification with **answer generation** using **BioGPT**. In-

puts are formatted as structured prompts combining question and context:

```
Question: {question}
Context: {context}
Answer with one word: yes, no, or maybe.
Answer:
```

This prompt formulation restricts the model to generate one of three valid outputs (yes, no, maybe), aligning the generative task with the classification setting used in previous experiments. All inputs are tokenised using the BioGPT tokenizer with a maximum sequence length of 512 tokens and contexts exceeding this limit are truncated accordingly. We performed zero-shot inference with the BioGPT to provide a baseline without task-specific supervision, leveraging the model's pre-training on biomedical literature. This zero-shot setting achieved an accuracy of 34%.

To further improve performance, we used parameter-efficient-fine-tuning (PEFT) using LoRA on the training split of the PubMedQA labelled data set as in previous experiments. Instead of updating all the parameters of the model, LoRA helps us to inject trainable low rank matrices in the attention layer, which significantly reduces the computational cost of the fine-tuning process. We specifically configured LoRA with rank=8, scaling factor=16, and dropout=0.05. Low rank is chosen to reduce the trainable parameters and the dropout configuration helps us avoid overfitting. LoRA is applied to query, key and value modules of the transformer, while keeping the rest of the model frozen.

For fine-tuning, the prompt is extended to include the ground truth label (Answer: answer). During training, all prompt tokens are masked except the answer tokens (yes/no/maybe), ensuring that the model learns to predict the answer conditioned on the input rather than modelling the full input. This aligns generation with the three-class classification objective. The model is fine-tuned for three epochs using a batch size of two and with a learning rate of 2×10^{-4} .

To address the class imbalance, 'no' and 'maybe' classes are upsampled. Evaluation is performed on a held-out test set of 500 examples. The generated text outputs are mapped to the class labels by a case-insensitive string mapper, to avoid formatting variation from the prediction.

3.1.5 Experiment 4: Biomed-R1-8B

BioMed-R1-8B replaces BioGPT’s next-token generation objective with a supervised fine-tuning and reinforcement learning pipeline that explicitly rewards correct multi-step inference, encouraging the model to backtrack and self-correct rather than commit to the first plausible token.

The experimental setup used the same 500/500 train-test split used in previous experiments. Due to GPU constraints of large models, parameter-efficient fine-tuning was applied via QLoRA. The model was fine-tuned using the SFTTrainer with an effective batch size of 8. A structured abstract format was used as input, where the question and abstract are concatenated into a single input string preserving section headings, and the model was trained to conclude responses with an explicit Final Answer: label token sequence.

3.2 Results

Overall, the results show that the best model depends on the evaluation objective (3). BioMed-R1-8B maximised headline accuracy at 0.782, with strong yes and no performance ($F1_{\text{yes}} = 0.860$; $F1_{\text{no}} = 0.790$), but failed completely on the minority maybe class ($F1_{\text{maybe}} = 0.000$). BioClinical ModernBERT provided the strongest efficient discriminative classifier, achieving 0.702 accuracy and 0.560 macro-F1, with particularly strong no-class performance ($F1_{\text{no}} = 0.727$). BioGPT achieved lower accuracy at 0.674, but matched BioClinical ModernBERT on macro-F1 (0.560) and was most effective at preserving uncertainty, achieving the highest maybe-class score ($F1_{\text{maybe}} = 0.410$).

3.3 Error Analysis

3.3.1 Baseline

The results from the baseline system indicate a heavily skewed and weakly discriminative system, achieving moderate overall accuracy (**0.516**) with low macro F1 (**0.353**). Performance is dominated by the ‘yes’ class ($F1=0.643$) while ‘no’ and ‘maybe’ are substantially weaker ($F1=0.339$, 0.097). Error analysis indicates a systematic bias towards assigning high confidence ‘yes’ labels even for clearly negative or uncertain cases, suggesting that the model is overfitting to lexical or semantic signals correlated with affirmative outcomes rather than capturing true decision boundaries. The highest-confidence errors further highlight poor calibration, as incorrect predictions exceed 0.75 prob-

ability, indicating overconfident misclassification. Mislabelling of ‘maybe’ instances as ‘yes’ suggests that the model fails to capture uncertain language, likely due to the limitations of TF-IDF features in representing nuanced scientific discourse. Overall, the baseline is biased and insensitive to minority or ambiguous classes, implying that improvements will require better handling of class imbalance, richer semantic representations, and explicit modelling of uncertainty.

3.3.2 Experiment 1: BiomedBERT Embeddings and Logistic Regression

Compared to the baseline system, Experiment 1 showed marginally improved accuracy (**0.532**) and increased macro-F1 (**0.438**). The most prominent error was misclassification of ‘no’ instances as ‘yes’, indicating that the model struggles to capture negations and instead relies on surface-level lexical similarity. Additionally, the ‘maybe’ class was poorly modelled with most instances being incorrectly assigned, suggesting an inability to detect uncertainty in the language. A clear bias towards predicting ‘yes’ class was identified, reflecting overconfidence in positive evidence. Finally, the model exhibited substantial confusion between ‘yes’ and ‘no’ highlighting that the model fails to distinguish between supporting and contradictory evidence, likely due to the limitations of a linear classifier operating on fixed embeddings. Further limitations include the ‘frozen’ nature whereby representations are not adapted to this specific classification task.

Despite these limitations, BiomedBERT with diff combination consistently outperforms the TF-IDF baseline across all metrics, demonstrating the value of domain-specific contextual representations even in a frozen, non-fine-tuned setting. These findings motivate the use of end-to-end fine-tuning in subsequent experiments.

3.3.3 Experiment 2: BioClinical ModernBERT

Relative to Experiment 1, Experiment 2 produced a clear improvement in balanced biomedical question answering using end-to-end fine-tuned BioClinical ModernBERT. The simple supervised Pipeline-S achieved the stronger mean macro F1 (**0.581** vs **0.578**), while the advanced semi-supervised Pipeline-E achieved higher accuracy (**0.740** vs **0.712**) and higher $F1_{\text{yes}}$ (**0.821** vs **0.795**). This demonstrates a key trade-off: pseudo-

	Accuracy	Macro F1	F1 _{yes}	F1 _{no}	F1 _{maybe}
Baseline (TF-IDF + LR)	0.516	0.353	0.643	0.339	0.097
Experiment 1 (BiomedBERT + LR)	0.532	0.438	0.637	0.462	0.214
Experiment 2 (BioClinical ModernBERT)	0.702	0.56	0.799	0.727	0.143
Experiment 3 (BioGPT)	0.674	0.56	0.77	0.50	0.41
Experiment 4 (Biomed-R1-8B)	0.782	0.5483	0.86	0.79	0.00

Table 3: Results comparing the Baseline system vs Experiments across: Accuracy, Macro-F1 and per-class F1 scores.

labelling improved majority-class performance, but not the balanced three-class objective that mattered most. The higher macro-F1 performance is attributed to increased cross-fold stability of the simple Pipeline-S. The advanced system had a higher ceiling but also a lower floor. Its main weakness was pseudo-label quality: when class-aware pseudo-labelling recovered enough clean minority-class candidates it performed well, but when those candidates collapsed the student mainly inherited extra confidence on *yes* rather than better balanced reasoning. Detailed pseudo-label and temperature diagnostics are reported in Appendix Tables 6 and 7. Calibration was justified in both pipelines as learned temperatures were frequently greater than 1, showing that raw probabilities were generally overconfident and needed to be softened before thresholding and ensembling. This mattered especially in the advanced pipeline, where pseudo-label acceptance depended directly on calibrated confidence and margin thresholds.

After applying the pre-registered selection rule, the simple supervised Pipeline-S was deployed as a calibrated 10-member soft-voting committee on the untouched 500-example test set, achieving an accuracy of **0.702** and a macro F1 of **0.56**. Relative to the development-side mean of **0.581** macro F1, this is a modest drop under stricter unseen evaluation rather than a collapse in generalisation.

Error analysis indicates weak handling of ‘maybe’ instances with only 8/55 being correctly classified. This weakness is identified not as a broad biomedical misunderstanding but failure to preserve ambiguity once evidence became moderately directional. In application terms, the model is strongest on clear biomedical conclusions and weakest on abstracts where evidence is mixed, in-

conclusive, or only partially supportive. Overall, Experiment 2 demonstrated clear improvement using end-to-end BioClinical ModernBERT but that the main remaining bottleneck is still **minority-class uncertainty modelling**, especially on ‘maybe’ instances.

3.3.4 Experiment 3: BioGPT

Experiment 3 did not demonstrate expected performance gain inline with previous experiments but did show increased F1_{maybe} indicating better handling of ‘maybe’ class than other experiments. This can be attributed to prompt engineering which explicitly instructed the model to output ‘maybe’ for weak or contradictory evidence. This improved class balance but significantly reduced accuracy (**0.674**) by biasing predictions. The inherent mismatch between the generative nature of BioGPT and the classification objective of this task provides an additional challenge. BioGPT would likely demonstrate increased performance without the constrained output prompting. Class-wise results showed strong performance on *yes* (F1_{yes} = **0.77**), moderate performance on *no* (F1_{no} = **0.50**), and weaker performance on *maybe* (F1_{maybe} = **0.41**), consistent with previously observed challenges of class imbalance. An additional challenge was the relatively small size of the labelled dataset which limited the model’s ability to learn robust decision boundaries. Input length constraints (maximum 512 tokens) further restrict performance, as longer contexts must be truncated, potentially removing important information. Finally, the use of parameter-efficient fine-tuning via LoRA, while computationally efficient, limits the extent to which the model can adapt compared to full fine-tuning, which has been shown to yield stronger perfor-

mance in prior work (Luo et al., 2022).

3.3.5 Experiment 4: Biomed-R1-8B

Biomed-R1-8B demonstrated the highest accuracy of **0.782** with a macro F1 of **0.548**. Compared to BioGPT, which demonstrated marginal 'maybe' predictions, Biomed-R1-8B showed complete failure on this class ($F1_{\text{maybe}}=0.00$). All 55 gold 'maybe' samples in the test were misclassified, 30 as 'yes' and 25 as 'no' meaning the model's effective output distribution was entirely binary. This is primarily caused by the under-representation of the 'maybe' class in the dataset (11%) which prevents adequate exposure of this class and eventually biases the decision of the model towards a definitive binary yes/no. A further identified error is the failure of sequence truncation. Fine-tuning process of the model was capped at 512 tokens to fit T4 VRAM constraints, yet many PubMedQA abstracts particularly those with detailed Results or Conclusions sections exceed this limit.

3.4 Answer Quality Prediction from Question Characteristics

To further understand key determinants of answer quality we analysed our final results across all models to understand the effect of question characteristics on output correctness. We averaged correctness across multiple QA system variants to obtain a continuous difficulty score, mitigating bias from any single pipeline configuration. This enables analysis of intrinsic question properties, including length, linguistic structure and reasoning requirements. Performance was evaluated using mean squared error (MSE) and Pearson correlation (r), capturing absolute prediction error and alignment between predicted and true difficulty trends.

Across the models tested, simple linear approaches proved the most effective. Lasso regression achieved the best performance ($\text{MSE } 0.076 \pm 0.009$, $\text{Pearson } r \ 0.603 \pm 0.056$), outperforming Ridge and Bayesian Ridge while maintaining stronger generalisation than the multilayer perceptron (MLP). The relatively poor performance of the neural model likely reflects the limited dataset, where higher-capacity non-linear models are more prone to overfitting. Lasso feature analysis showed that reasoning-related annotations ('reasoning-free'/'multi-step reasoning' derived from the PubMedQA dataset) were the strongest predictors of answer quality. Overall, these findings indicate that the cognitive com-

plexity of a question is the primary determinant of QA performance, outweighing more surface-level characteristics such as length, vocabulary, or entity count, which emphasises the necessity of reasoning-capable models.

3.5 Discussion

Overall, our results show that task-specific improvements across both text representation and prediction method improved performance. Along the representation axis, performance improved as we moved from sparse lexical features to biomedical contextual representations. The TF-IDF + Logistic Regression baseline was simple and interpretable, but struggled with negation, uncertainty and semantic relationships. Replacing TF-IDF with frozen BiomedBERT embeddings improved balanced performance, showing that domain-specific contextual representations were useful even without full fine-tuning. However, the frozen encoder and linear classifier still had limited ability to adapt to PubMedQA-specific decision boundaries. BioClinical ModernBERT addressed this by learning the representation and classifier jointly, producing the strongest discriminative classifier result with 0.702 accuracy and a final macro-F1 of **0.560** on the untouched test set.

Along the prediction-method axis, the results were more nuanced. End-to-end classification was more reliable than linear classification, but greater model complexity did not always improve balanced performance. Within BioClinical ModernBERT, the semi-supervised Pipeline-E achieved higher accuracy and stronger yes' class performance, while the simpler supervised Pipeline-S achieved slightly higher macro-F1 and greater stability. This suggests that pseudo-labelling mainly strengthened majority-class confidence rather than resolving uncertainty. BioGPT showed a different trade-off: it did not outperform BioClinical ModernBERT on headline accuracy, achieving 0.674 compared with 0.702, but matched it on macro-F1 (0.560) and achieved the strongest maybe' performance, with $F1_{\text{maybe}} = 0.410$ compared with 0.143 for BioClinical ModernBERT and 0.000 for BioMed-R1-8B. This suggests that constrained generative prompting was less effective for maximising headline accuracy, but more effective at preserving uncertainty in inconclusive cases.

This trade-off was clearest for BioMed-R1-8B. It achieved the highest headline accuracy at 0.782, with strong 'yes' and 'no' performance, but col-

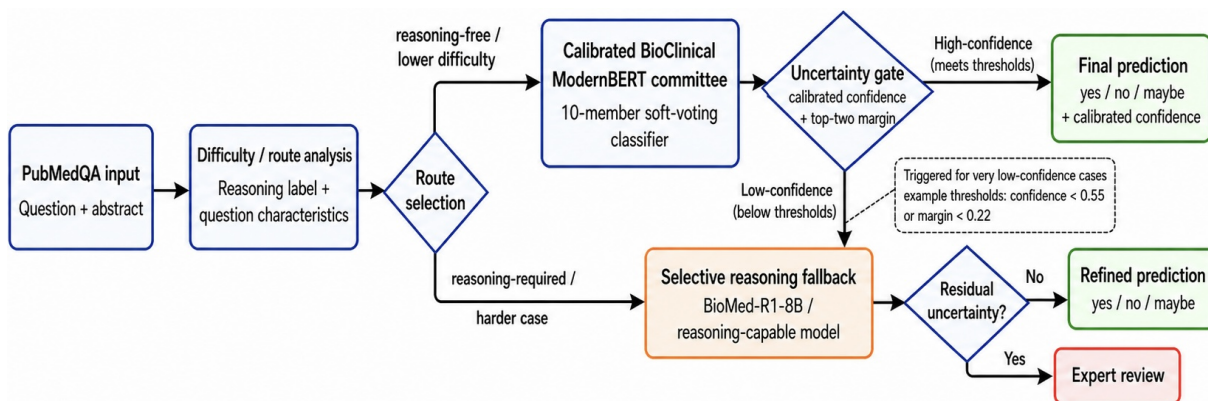


Figure 4: Proposed Uncertainty-aware cascade for PubMedQA.

lapsed completely on the ‘maybe’ class. This makes it unsuitable as a standalone biomedical QA system, despite its strong accuracy. In PubMedQA, preserving uncertainty is important because biomedical studies often produce mixed, limited or inconclusive evidence. A model that forces these cases into binary ‘yes’ or ‘no’ answers risks overstating the certainty of the evidence.

The question-characteristic analysis further supported this interpretation, showing that reasoning-related features were key determinants of answer quality. This suggests that a single-model system is unlikely to be optimal: easier, high-confidence cases can be handled efficiently by a calibrated classifier, while reasoning-heavy or ambiguous cases require additional processing. Figure 4 therefore proposes an uncertainty-aware cascade based on the main failure modes observed in our experiments. The BioClinical ModernBERT committee first produces calibrated probabilities; cases with high confidence and a clear top-two margin (between class probabilities) are answered directly, while low-confidence or reasoning-heavy cases are selectively escalated to a generative or reasoning-capable model. Cases that remain uncertain after escalation are referred for expert review rather than forced into an answer.

This design follows directly from the observed trade-off between models. BioClinical ModernBERT provides efficient classification, BioGPT is better at preserving uncertainty, and BioMed-R1-8B achieves strong yes/no accuracy but should not be used alone because it collapses on the ‘maybe’ class. An uncertainty-aware cascade would therefore combine the efficiency of smaller classifiers with the reasoning capacity of larger models, while preserving uncertainty where it matters most in

biomedical evidence interpretation.

4 Conclusion

This project investigated biomedical question answering on PubMedQA by varying two central parts of the pipeline: **text representation** and **prediction method**. We found that both axes affected performance, but improvements were strongest when they addressed specific task bottlenecks such as biomedical semantics, class imbalance, uncertainty and reasoning. PubMedQA consists of dense biomedical abstracts, domain-specific language and class imbalance, requiring models that can capture meaning, evidence direction and uncertainty. Our strongest discriminative classifier was BioClinical ModernBERT, which achieved an accuracy of 0.702, while BioMed-R1-8B achieved the highest headline accuracy at 0.782. However, our results also show that accuracy alone is not sufficient for judging answer quality. BioMed-R1-8B failed completely on the minority maybe class ($F1_{\text{maybe}} = 0.000$), whereas BioGPT, despite lower accuracy at 0.674, achieved the strongest maybe-class performance ($F1_{\text{maybe}} = 0.410$) and matched BioClinical ModernBERT on macro-F1 (0.560). This suggests that generative prompting may be useful for preserving uncertainty, even when it does not maximise overall accuracy. Future work should therefore not simply scale model complexity, but combine the strengths of different systems through an uncertainty-aware cascade: a calibrated classifier for efficient high-confidence cases, a generative or reasoning model for ambiguous and reasoning-heavy cases, and expert review where uncertainty remains high.

¹

¹GitHub link: <https://github.com/emilyrbee/Group-21-.git>

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5 BioClinical ModernBERT Appendix

Table 4 records the key preprocessing and Stage 1 optimisation settings that supported the main BioClinical ModernBERT pipeline. Table 5 records the detailed Stage 2 semi-supervised settings used for pseudo-label selection, student training, and the optional reasoning fallback. Table 7 summarises the most important supporting diagnostics behind the final comparison, and Table 8 gives the final official test confusion matrix for the selected calibrated committee.

Quantity	Value
Labels	no=0, maybe=1, yes=2
Input format	QUESTION + CONTEXTS
Truncation	only_second
Max length	512
Mean length	302.08
Median length	300
95th percentile	443.05
99th percentile	538.02
Examples > 512	90 / 5000 (1.80%)
Backbone	BioClinical ModernBERT
Best imbalance method	weighted sampler only
Class-weighted loss	no
Supervised learning rate	8×10^{-6}
Gradient accumulation	2
Padding	dynamic
Model selection	validation macro F1
Weak-stage labels	binary yes/no
Weak-stage pilot split	10% subset
Weak-stage best learning rate	1×10^{-5}
Weak-stage exported output	encoder only

Table 4: Supporting preprocessing and Stage 1 settings for BioClinical ModernBERT.

Quantity	Value
Teacher agreement required	yes
Two-teacher purpose	diversity for agreement-based pseudo-label filtering
Confidence threshold (no, yes)	0.78
Margin threshold (no, yes)	0.12
Confidence threshold (maybe)	0.62
Margin threshold (maybe)	0.05
Pseudo-label multiplier	$2.5 \times$ training fold
Typical pseudo-label target (i.e. number of pseudo-labels per student fold)	1028
Typical pseudo-label target (no)	338
Typical pseudo-label target (maybe)	138
Typical pseudo-label target (yes)	552
Maybe boost	25%
Pseudo-label type	soft labels
Pseudo-label loss weight	0.35
Student learning rate	6×10^{-6}
Student epochs	up to 6
Student weighted sampling (off as it comes with own method of handling maybe imbalance)	off
Calibration purpose	soften overconfident probabilities before thresholding and ensembling
Final committee size	10
Committee voting rule	soft voting to preserve calibrated probabilities and reduce variance
Selective fallback trigger: confidence	< 0.55
Selective fallback trigger: top-two margin	< 0.22
Selective fallback budget	20% of test set
Reasoning model	BioMed-R1-8B
Fallback purpose	escalate only very low-confidence committee cases to reasoning model
Decoding	deterministic
Max new tokens	128
Prompt max length	2048
Few-shot examples in prompt	3 (one per class)
Fallback if parse fails	original committee prediction

Table 5: Supporting Stage 2 semi-supervised and post-selection settings for BioClinical ModernBERT.

Table 6: Upstream route-selection summary for Experiment 2. The masked-adaptation route was rejected because it did not improve downstream three-class QA. The binary weak-label stage uses only *yes/no*, so *maybe* is not applicable there.

Stage / system	Accuracy	Macro F1	F1 No	F1 Maybe	F1 Yes
Supervised base route (3-seed mean)	0.493	0.476	0.366	0.492	0.568
MLM-adapted route (pilot downstream check)	0.480	0.453	0.333	0.444	0.582
Full weak PQA-A warm-start (binary)	0.981	0.922	0.854	–	0.990

Quantity	Pipeline S	Pipeline E
Mean outer accuracy	0.712 ± 0.038	0.740 ± 0.044
Mean outer macro F1	0.581 ± 0.050	0.578 ± 0.076
Mean outer $F1_{\text{maybe}}$	0.208 ± 0.129	0.185 ± 0.160
Best fold macro F1	0.680 (fold 7)	0.702 (fold 2)
Simple-pipeline temperature range	0.762–1.981	–
Final committee temperature range	0.948–2.152	–
Selected pseudo-label count	–	578–1028
Key pseudo-label failure	–	fold 3: 0 <i>no</i> , 26 <i>maybe</i> candidates

Table 7: Compact supporting diagnostics for BioClinical ModernBERT.

True class	Pred no	Pred maybe	Pred yes
No	120	21	28
Maybe	16	8	31
Yes	25	28	223

Table 8: Final test confusion matrix for the Pipeline S calibrated committee. Rows are true labels and columns are predicted labels.

5.1 BioClinical ModernBERT training schematics

Figures 5a and 5b summarise the final BioClinical ModernBERT training design. The Stage 1 schematic shows the route-selection process: masked adaptation on unlabelled data was tested but rejected, and the stronger base classifier route was then warm-started on the PQA-A weak-label set. The Stage 2 schematic shows the standardised 500/500 expert split, the outer 10-fold cross-validation protocol, the comparison between the simple supervised and advanced semi-supervised pipelines, and the final calibrated committee evaluation on the untouched test set.

Step	Modification	Macro-F1	Main lesson
Base ModernBERT	Plain fine-tuning	0.41	Initial Supervised Baseline
+ weighted sampler	Imbalance handling	0.476	Improved minority-class exposure and became the stronger supervised route.
+ weak warm-start	PQA-A yes/no pretraining	0.922 [†]	Learned a strong binary biomedical yes/no prior before three-class expert fine-tuning.
+ calibration	Pipeline-S with temperature scaling	0.581	Produced the strongest and most stable outer-fold macro-F1.
+ pseudo-labelling	Pipeline-E	0.578	Improved accuracy and $F1_{yes}$, but did not improve macro-F1 over Pipeline-S.
Selected final route	Pipeline-S committee	0.56	Maintained performance on test set

Table 9: Ablation summary for BioClinical ModernBERT - Experiment 2. [†]The weak warm-start score is from the binary PQA-A binary yes/no task. Pipeline-E and Pipeline-S report the outer-fold development mean; the selected final route reports performance on the untouched test set.



(a) Stage 1 route selection and weak warm-start.

(b) Stage 2 standardised 500/500 evaluation protocol.

Figure 5: BioClinical ModernBERT training schematics.